

Phantom Fingers: A Table-Grounded Haptic Interface for Virtual Enviroments

Jorge Samuel Garcia Alonzo¹, Iqui Balam Heredia-Marin¹, Melisa Orta Martinez¹

Abstract

- We present a finger mechanism using PHANTOM-inspired parallel four-bar kinematics with capstan transmission. Workspace coverage relative to human finger kinematics improves from 80.4% → 96.9% (index) and 8.9% → 96.7% (thumb).

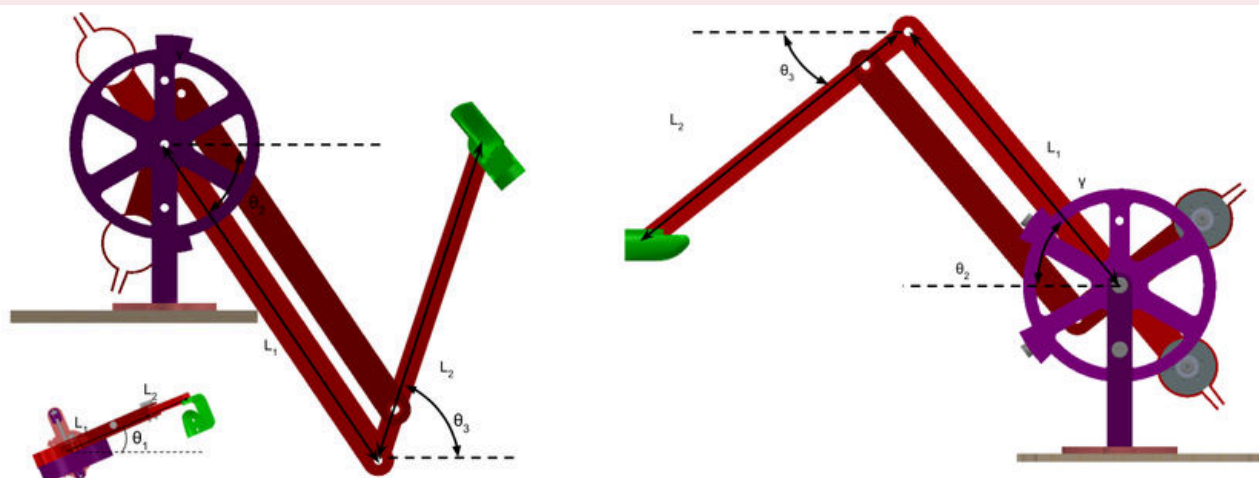
Background

Haptic interfaces enable intuitive control in remote manipulation and education, but low-cost designs struggle to match human finger kinematics. This ergonomic mismatch forces non-intuitive joint configurations to reach peripheral regions:

Workspace coverage vs. human reference [1][2]

- Index — 96.86%
- Middle — 97.14%
- Thumb — 96.65%

This work applies PHANTOM-inspired parallel four-bar kinematics [3] to a previous (Haplink/delta) low-cost haptic platform developed at CMU's SHRED Lab for undergraduate dexterity research.



$$x = s_1(l_1c_2 + l_2c_3), y = l_1s_2 - l_2s_3, z = c_1(l_1c_2 + l_2c_3)$$

$$x = l_1c_2 + l_2c_3, y = l_1s_2 - l_2s_3$$

$$\mathbf{J} = \begin{pmatrix} c_1(l_1c_2 + l_2c_3) & -s_1l_1s_2 & -s_1l_2s_3 \\ 0 & l_1c_2 & -l_2c_3 \\ -s_1(l_1c_2 + l_2c_3) & -c_1l_1s_2 & -c_1l_2s_3 \end{pmatrix}$$

$$\mathbf{J} = \begin{pmatrix} -l_1s_2 & -l_2s_3 \\ l_1c_2 & -l_2c_3 \end{pmatrix}$$

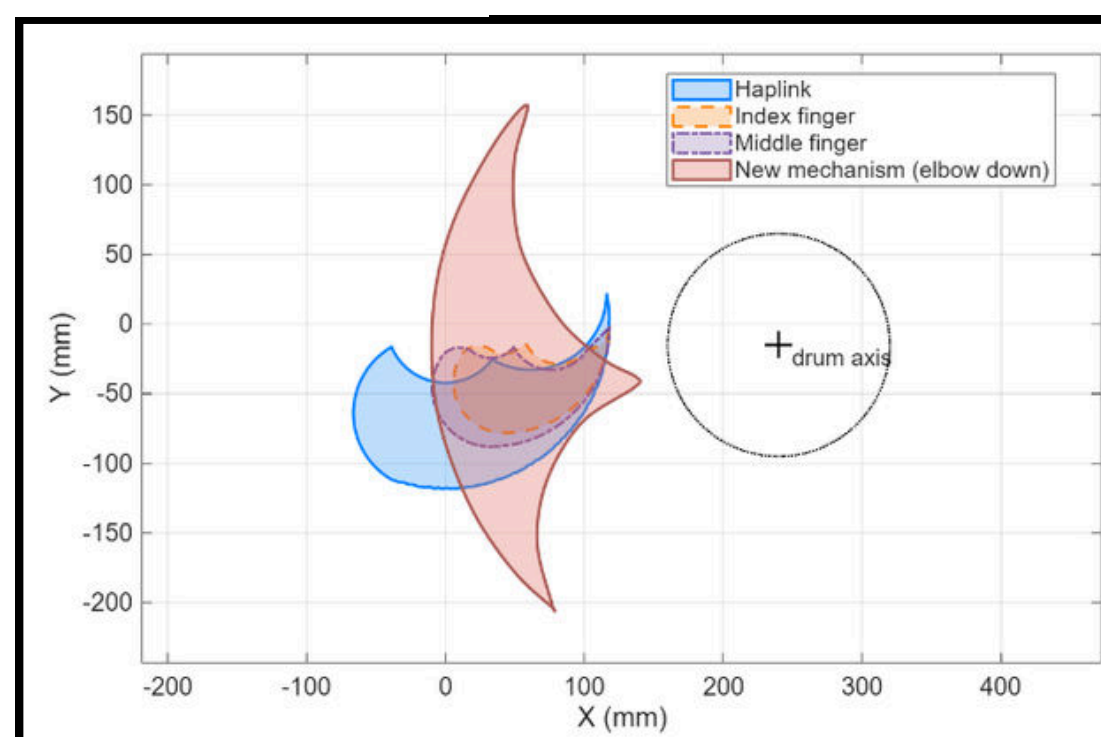


Fig. 1. Comparative workspace visualization of original and optimized DEXKIT mechanisms (2D projection) overlaid with human finger kinematic reach boundaries.

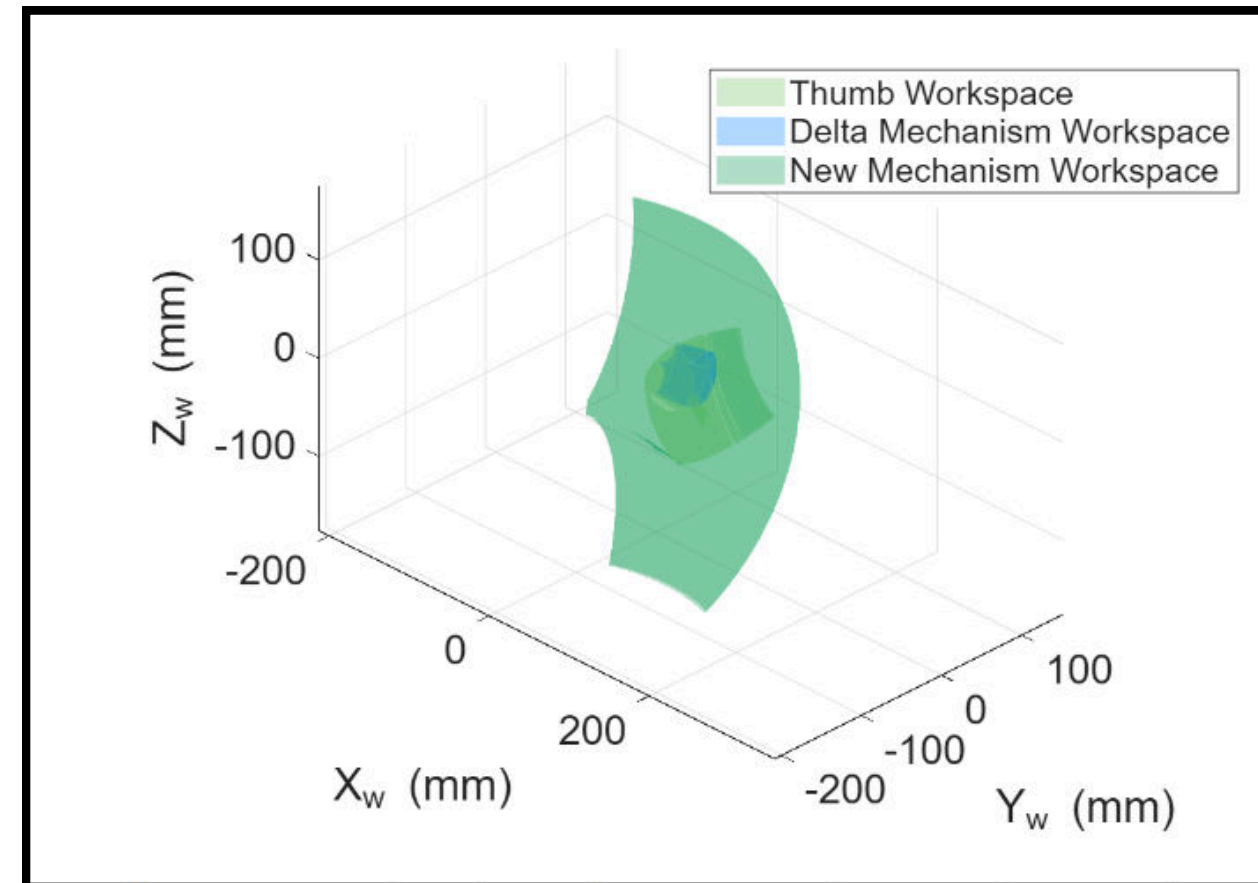


Fig. 2. 3D comparative workspace for thumb mechanism: Delta mechanism optimized PHANTOM-based design, and reference human thumb workspace.

Haptic Interface

Three independent capstan-driven modules, each a parallel four-bar linkage actuated by two brushless motors through a cable-driven drum. The 0.024" steel cable wraps three times per thread, giving a low-friction, backlash-free transmission at low cost. Structural parts are 3D-printed (FDM and SLA) with aluminum heat inserts at load-bearing threads, mounted on an aluminum-extrusion and MDF frame.

- Index, middle — mirrored linkages separated by the natural abduction angle, with free rotational bases for comfortable grasp closure; both share a unified platform
- Thumb — a dedicated third motor drives the rotational base ($\pm 20^\circ$) for out-of-plane opposition [1][2]. The elbow is inverted downward to avoid collision with the hand during closure, a failure mode of the previous prototype
- Control — real-time C++ kinematics; force rendering via $\mathbf{J}^T$, inverse kinematics via $\mathbf{J}^{-1}$, with 3D visualization for workspace validation

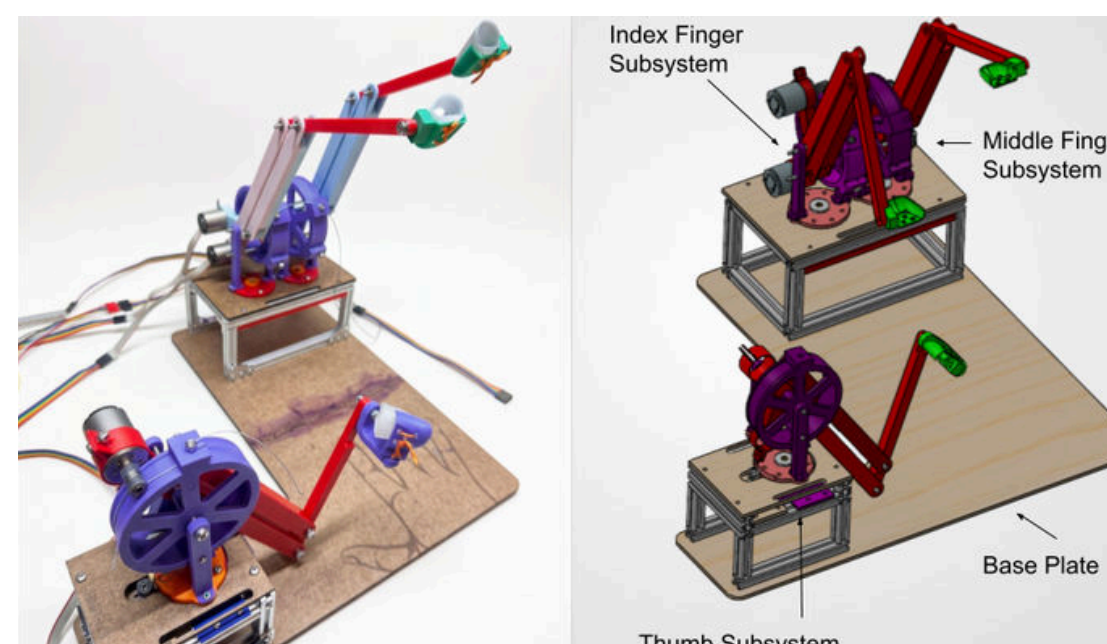


Fig. 3. CAD assembly of Phantom Fingers mechanism showing integrated finger modules with parallel four-bar linkage, motor-driven drum, and capstan cable transmission.

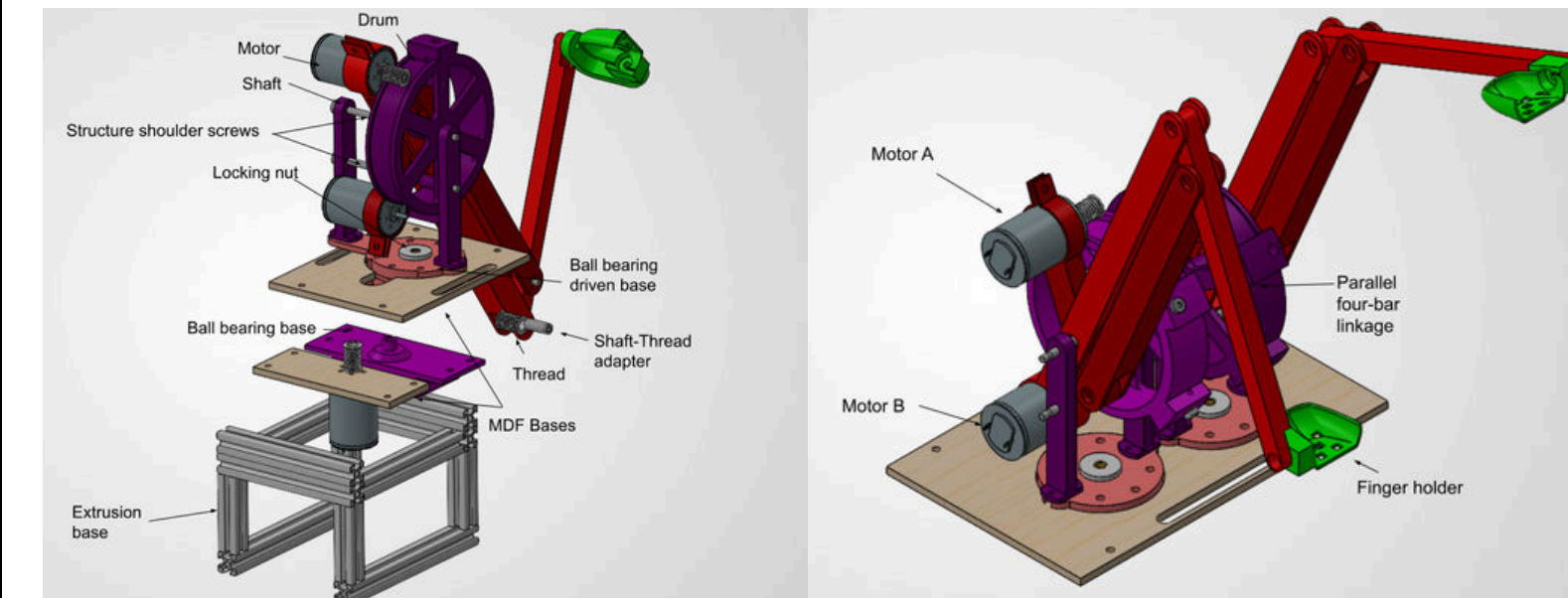


Fig. 4. CAD exploded view of the haptic thumb subsystem (left) and the index and middle finger subsystem (right).

What is next?

The next phase integrates the Phantom Fingers mechanism with a soft-foam pulley-actuated hand, transitioning from virtual simulation to physical tactile feedback delivery. This integration completes the DEXKIT project.

Current implementation lacks wrist sensing, limiting control authority during push-pull operations and reducing the fidelity of force feedback during manipulation. A sensor-equipped capstan wrist attachment would address this limitation by introducing a fourth degree of freedom while enabling real-time wrist torque measurement. This enhancement improves both visualization accuracy and ergonomic accessibility, providing a natural resting position during extended operation.

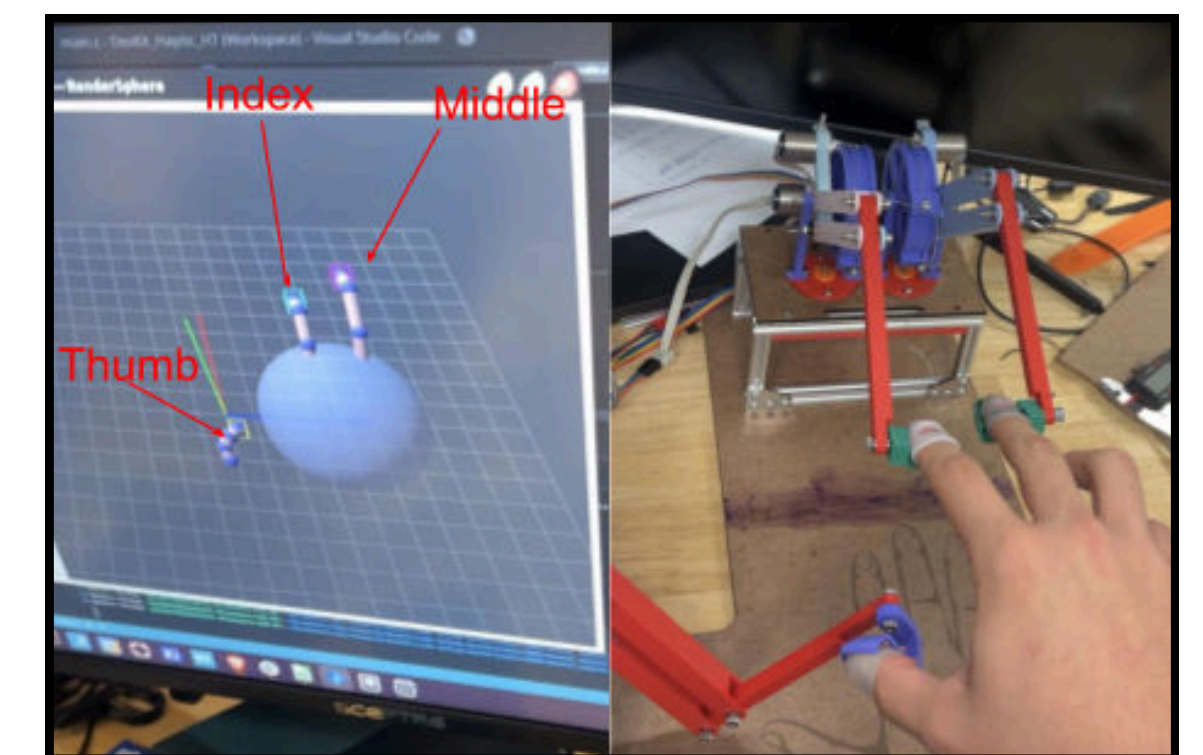


Fig.5. Complete prototype and visualization.

References

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- [2] Al-Qattan et al. (2021). J. Hand Surg. (Eur.), 46(10):1047–1056.
- [3] Brink et al. (2020). J. Hand Therapy, 34(1):32–41.
- [4] Buonamici et al. (2021). Appl. Ergonomics, 96:103487.
- [5] Kparanam & Madhavi (2019). Int. J. Eng. Res. Tech., 8(7):1–8.